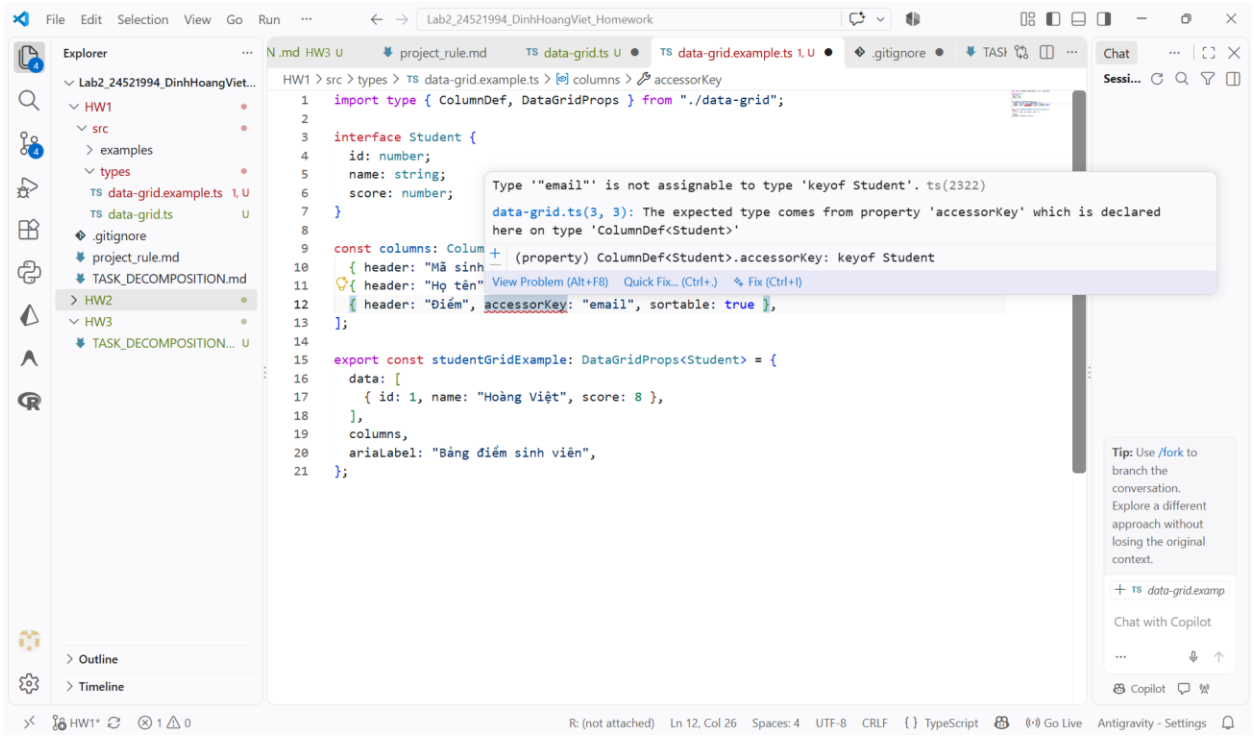
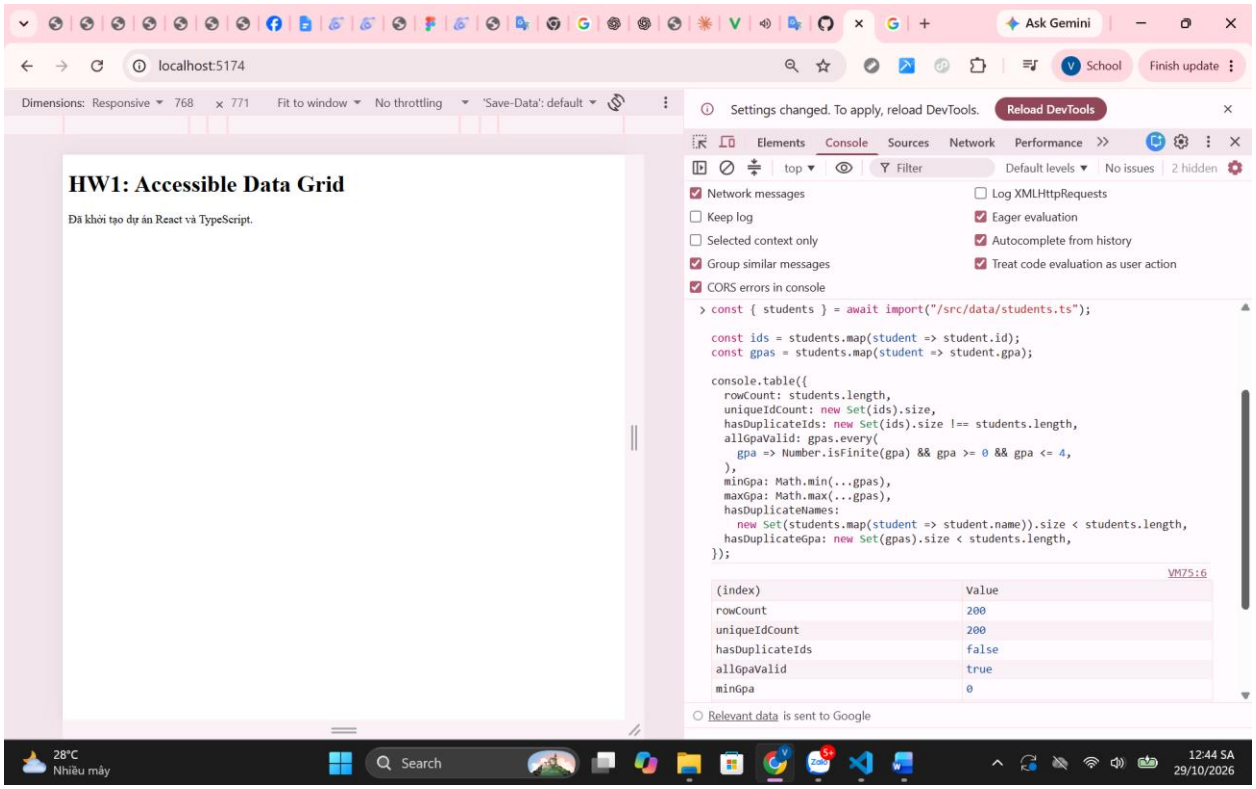
HW1:

1.3. When convert “score” into “email”



1.4. Configuration

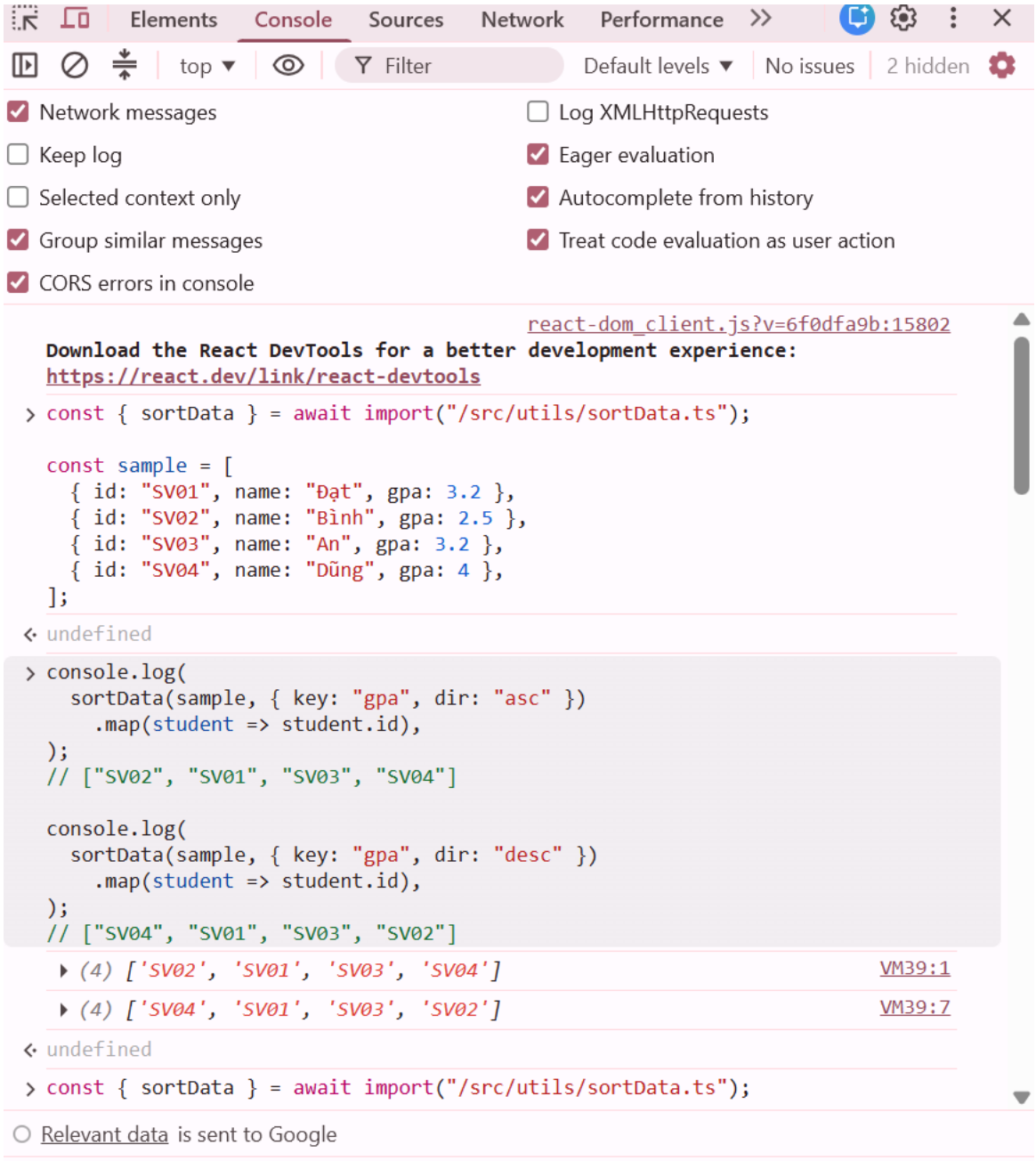


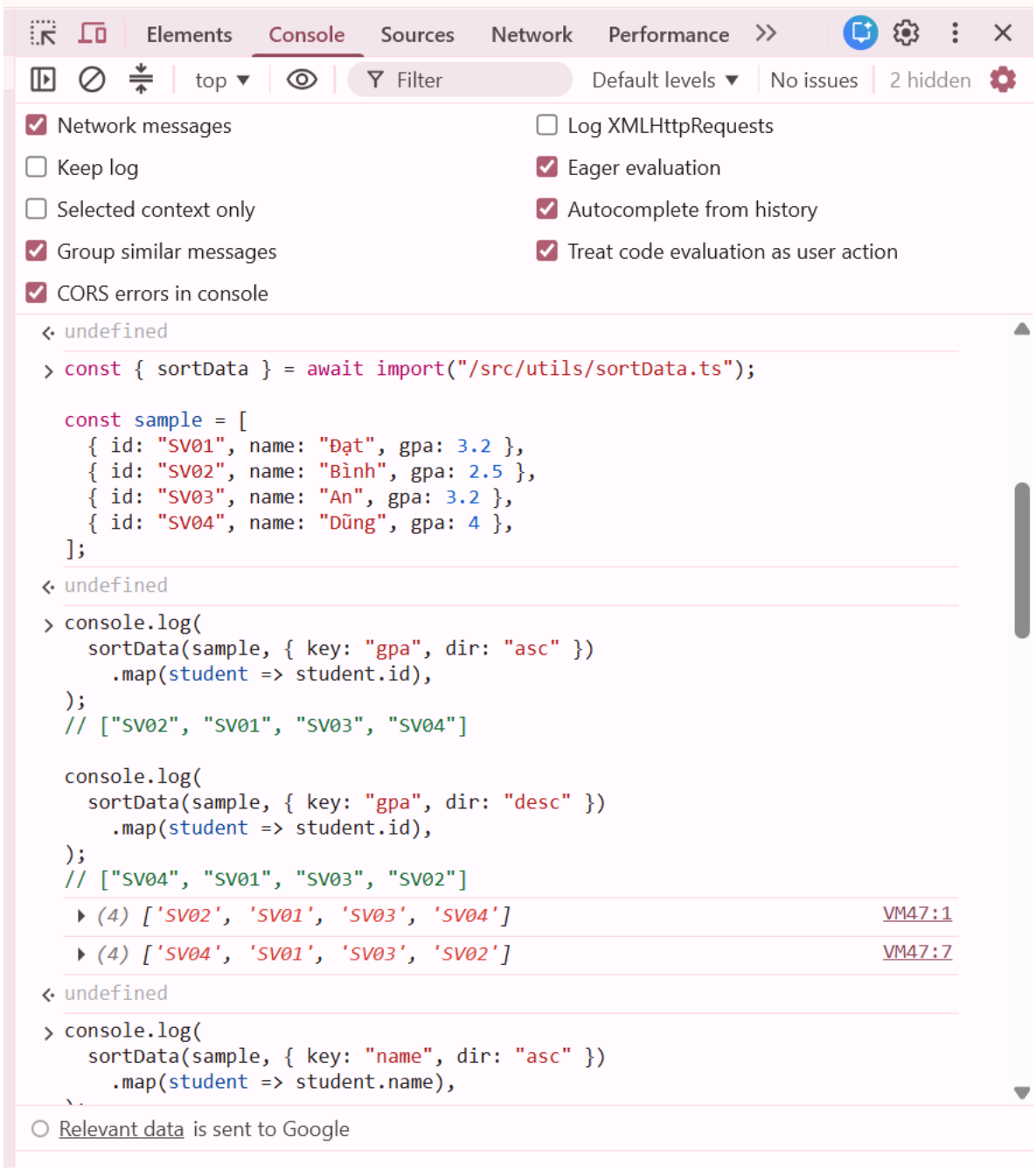
VM75:6

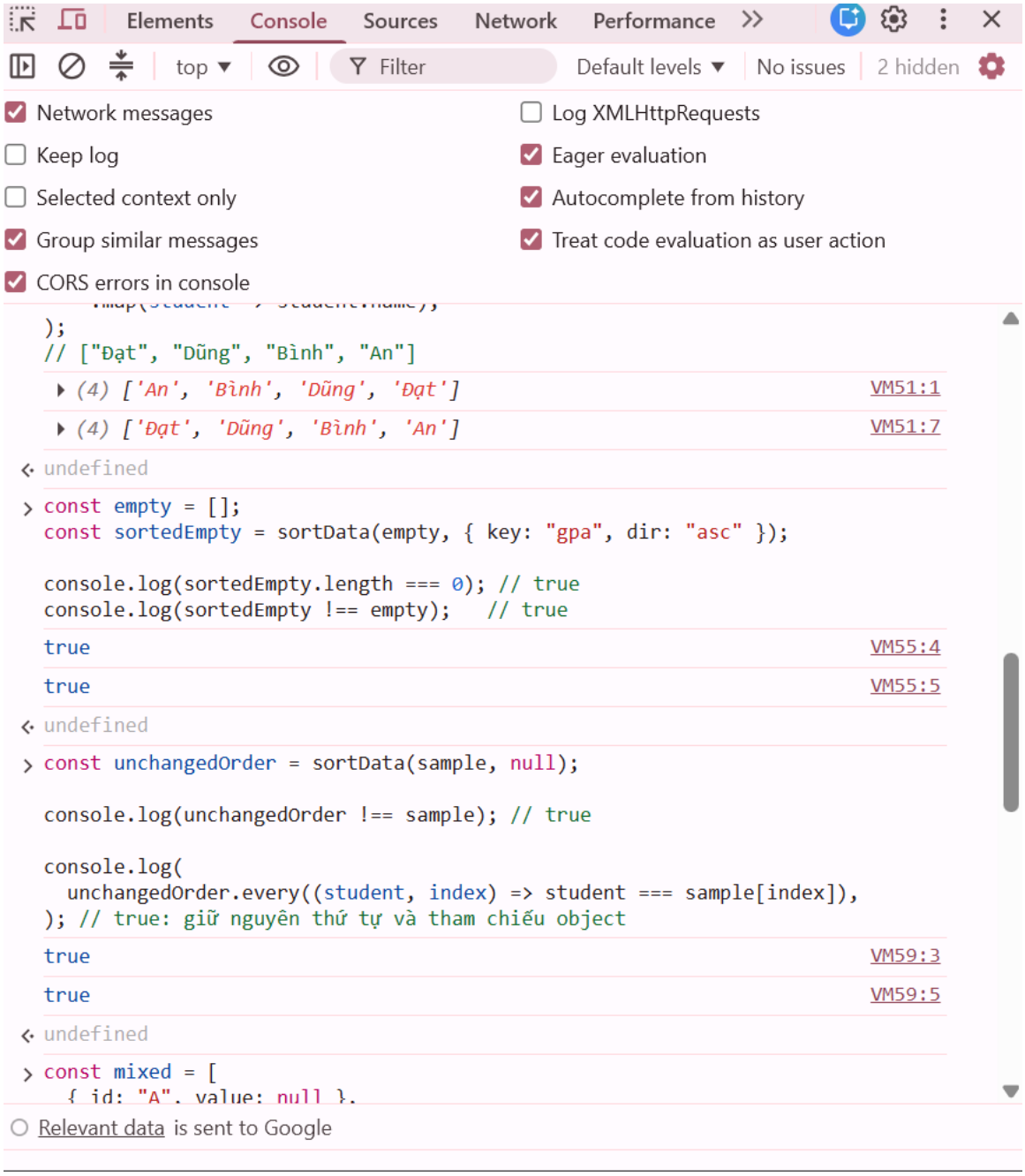
(index)	Value
rowCount	200
uniqueIdCount	200
hasDuplicateIds	false
allGpaValid	true
minGpa	0
maxGpa	4
hasDuplicateNames	true
hasDuplicateGpa	true

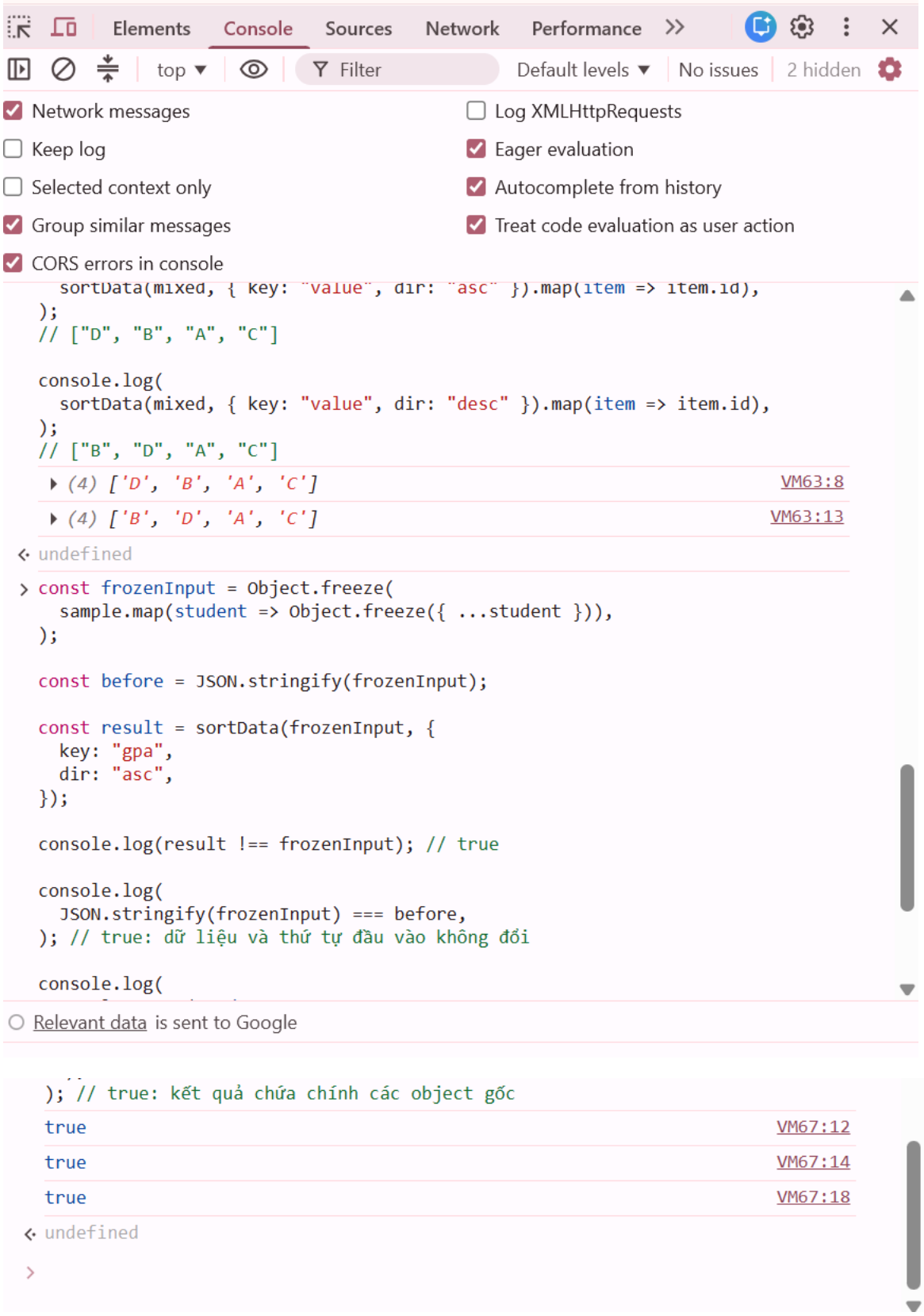
▸ Object

undefined

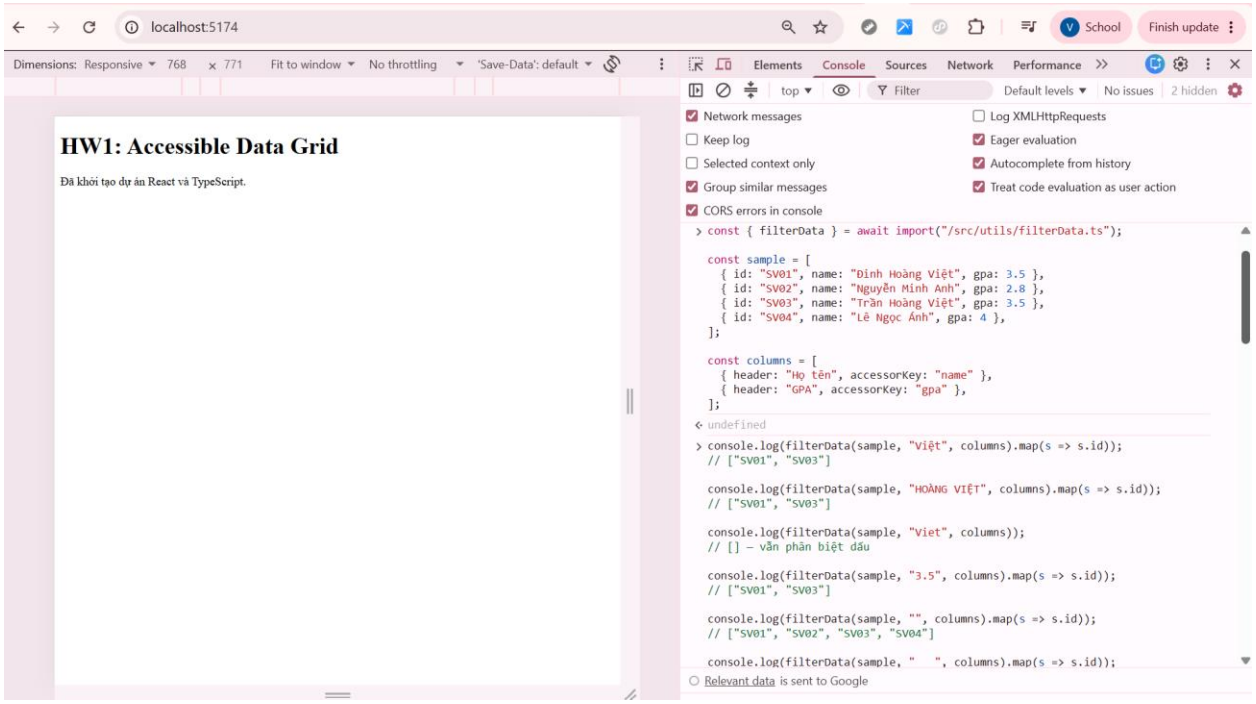

Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No issues
2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
react-dom_client.js?v=6f0dfa9b:15802
Download the React DevTools for a better development experience:
https://react.dev/link/react-devtools
> const { sortData } = await import("/src/utils/sortData.ts");
const sample = [
{ id: "SV01", name: "Đạt", gpa: 3.2 },
{ id: "SV02", name: "Bình", gpa: 2.5 },
{ id: "SV03", name: "An", gpa: 3.2 },
{ id: "SV04", name: "Dũng", gpa: 4 },
];
undefined
> console.log(
sortData(sample, { key: "gpa", dir: "asc" })
.map(student => student.id),
);
// ["SV02", "SV01", "SV03", "SV04"]
console.log(
sortData(sample, { key: "gpa", dir: "desc" })
.map(student => student.id),
);
// ["SV04", "SV01", "SV03", "SV02"]
(4) ['SV02', 'SV01', 'SV03', 'SV04']
VM39:1
(4) ['SV04', 'SV01', 'SV03', 'SV02']
VM39:7
undefined
> const { sortData } = await import("/src/utils/sortData.ts");
Relevant data is sent to Google

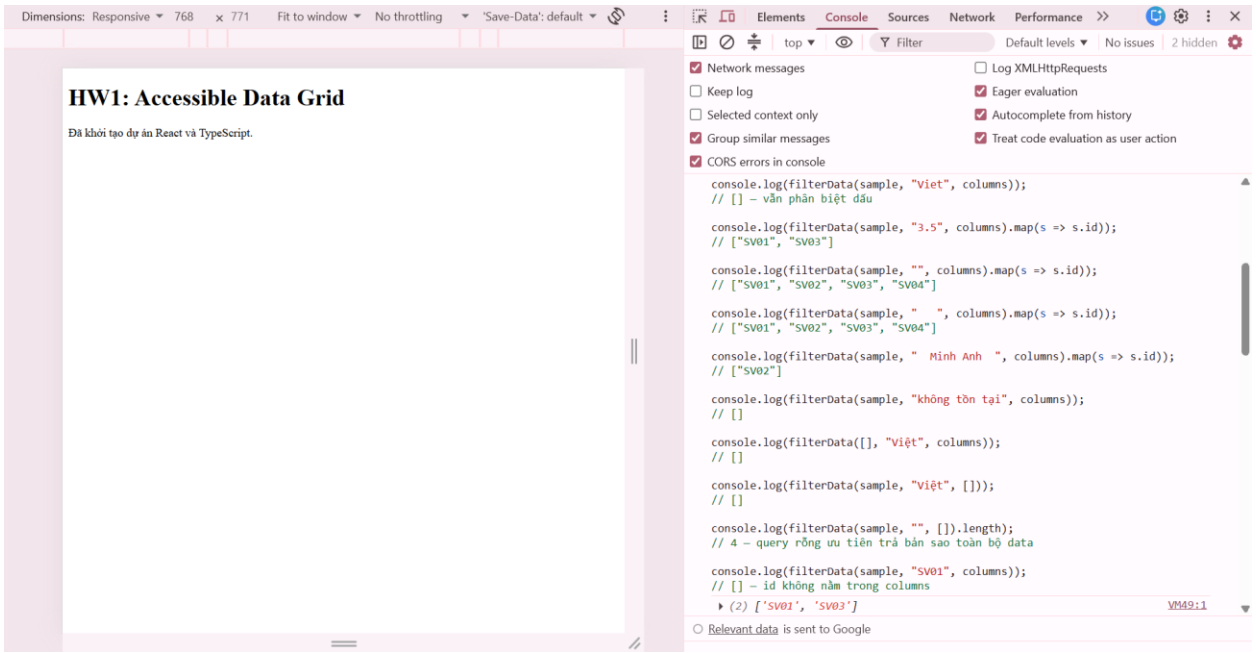

Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No issues
2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
undefined
const { sortData } = await import("/src/utils/sortData.ts");
const sample = [
{ id: "SV01", name: "Đạt", gpa: 3.2 },
{ id: "SV02", name: "Bình", gpa: 2.5 },
{ id: "SV03", name: "An", gpa: 3.2 },
{ id: "SV04", name: "Dũng", gpa: 4 },
];
undefined
console.log(
sortData(sample, { key: "gpa", dir: "asc" })
.map(student => student.id),
);
// ["SV02", "SV01", "SV03", "SV04"]
console.log(
sortData(sample, { key: "gpa", dir: "desc" })
.map(student => student.id),
);
// ["SV04", "SV01", "SV03", "SV02"]
(4) ['SV02', 'SV01', 'SV03', 'SV04']
VM47:1
(4) ['SV04', 'SV01', 'SV03', 'SV02']
VM47:7
undefined
console.log(
sortData(sample, { key: "name", dir: "asc" })
.map(student => student.name),
Relevant data is sent to Google

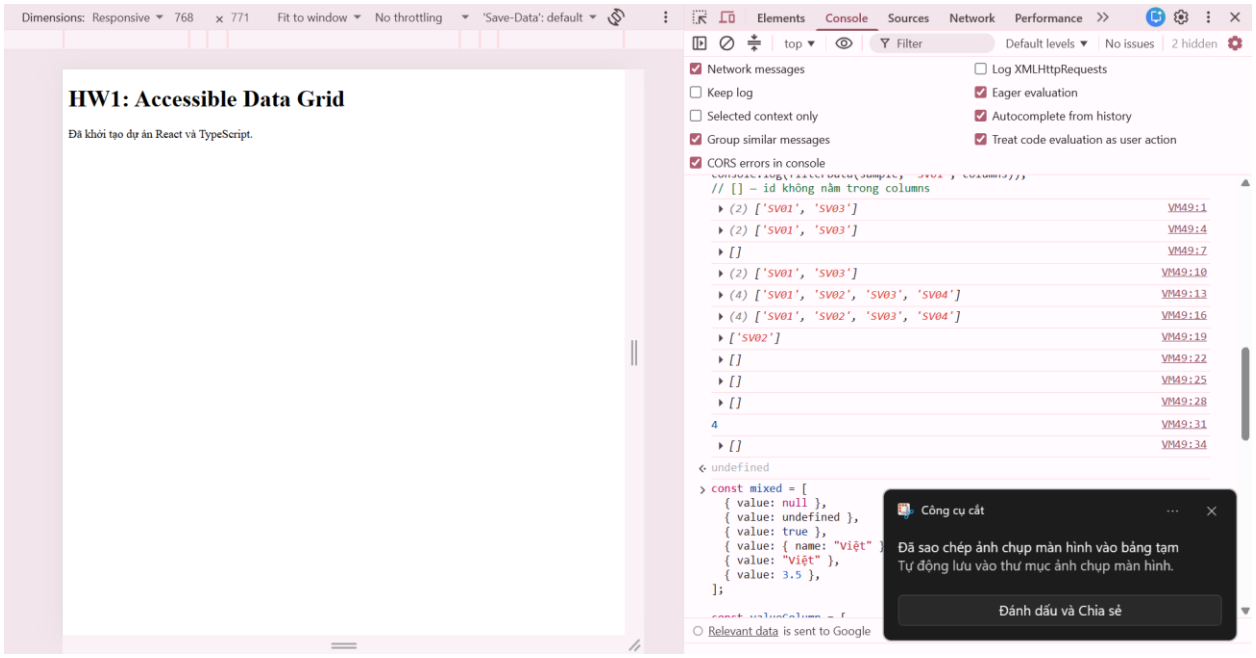

Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No issues
2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
);
// ["Đạt", "Dũng", "Bình", "An"]
(4) ['An', 'Bình', 'Dũng', 'Đạt'] VM51:1
(4) ['Đạt', 'Dũng', 'Bình', 'An'] VM51:7
undefined
const empty = [];
const sortedEmpty = sortData(empty, { key: "gpa", dir: "asc" });
console.log(sortedEmpty.length === 0); // true
console.log(sortedEmpty !== empty); // true
true VM55:4
true VM55:5
undefined
const unchangedOrder = sortData(sample, null);
console.log(unchangedOrder !== sample); // true
console.log(
unchangedOrder.every((student, index) => student === sample[index]),
); // true: giữ nguyên thứ tự và tham chiếu object
true VM59:3
true VM59:5
undefined
const mixed = [
{ id: "A", value: null },
Relevant data is sent to Google

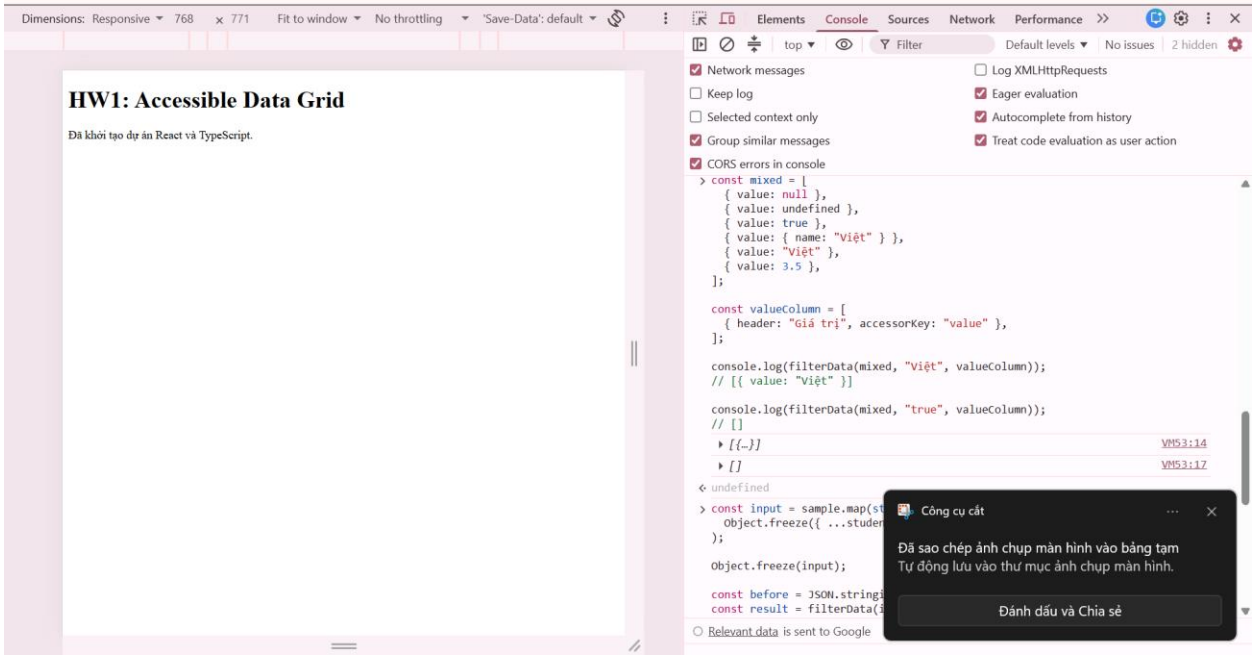


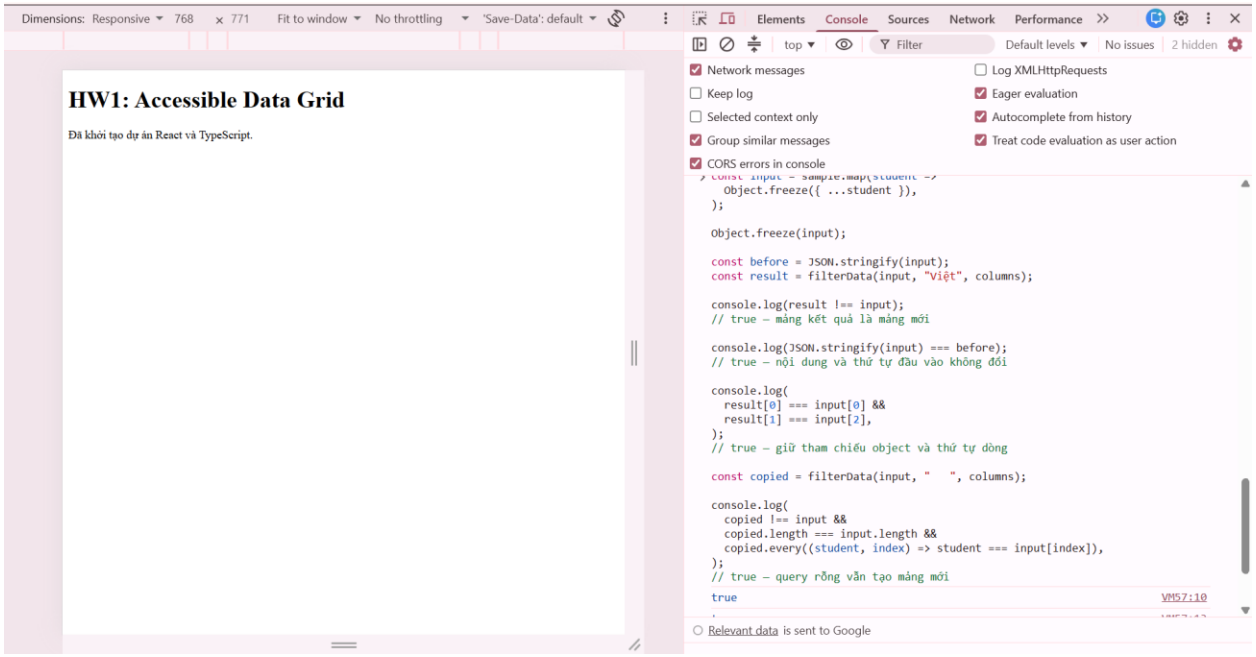
Task 2.2: filterData(data, query, columns)

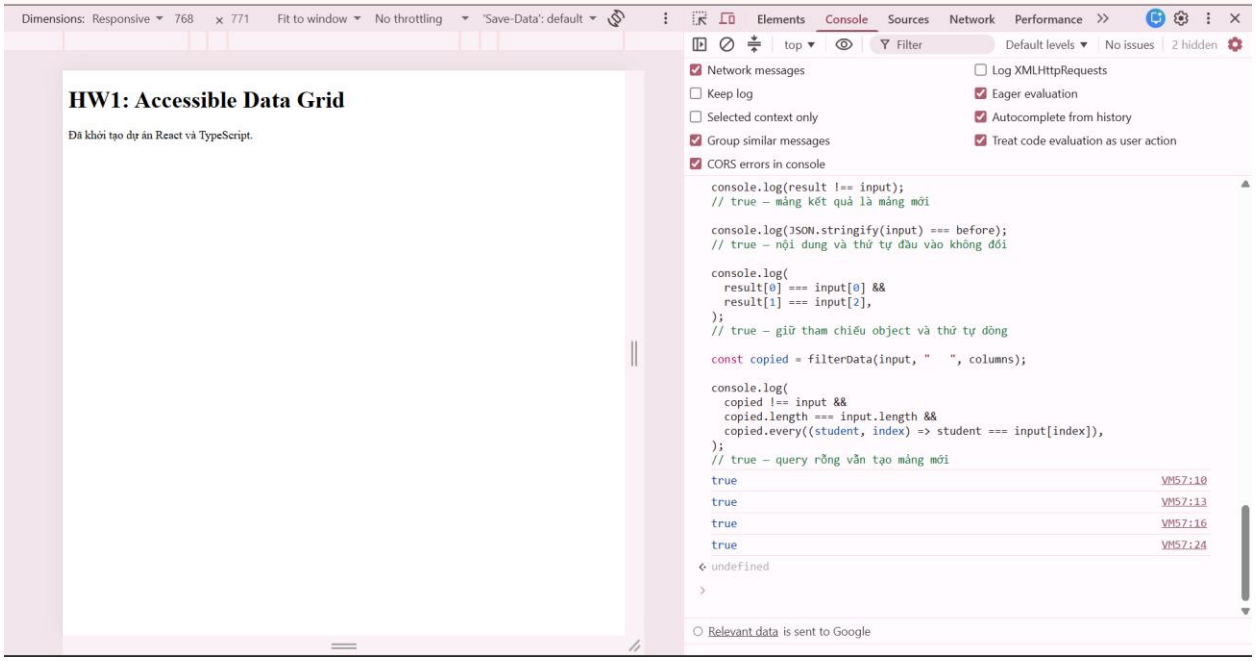

localhost:5174
School
Finish update
Dimensions: Responsive 768 x 771 Fit to window No throttling 'Save-Data': default
Elements Console Sources Network Performance
top Filter Default levels No issues 2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
HW1: Accessible Data Grid
Đã khởi tạo dự án React và TypeScript.
> const { filterData } = await import("/src/utils/filterData.ts");
const sample = [
{ id: "SV01", name: "Đinh Hoàng Việt", gpa: 3.5 },
{ id: "SV02", name: "Nguyễn Minh Anh", gpa: 2.8 },
{ id: "SV03", name: "Trần Hoàng Việt", gpa: 3.5 },
{ id: "SV04", name: "Lê Ngọc Ánh", gpa: 4 },
];
const columns = [
{ header: "Họ tên", accessorKey: "name" },
{ header: "GPA", accessorKey: "gpa" },
];
undefined
> console.log(filterData(sample, "Việt", columns).map(s => s.id));
// ["SV01", "SV03"]
console.log(filterData(sample, "HOÀNG VIỆT", columns).map(s => s.id));
// ["SV01", "SV03"]
console.log(filterData(sample, "Viet", columns));
// [] – vẫn phân biệt dấu
console.log(filterData(sample, "3.5", columns).map(s => s.id));
// ["SV01", "SV03"]
console.log(filterData(sample, "", columns).map(s => s.id));
// ["SV01", "SV02", "SV03", "SV04"]
Relevant data is sent to Google


Dimensions: Responsive 768 x 771 Fit to window No throttling 'Save-Data': default
Elements Console Sources Network Performance
top Filter Default levels No issues 2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
HW1: Accessible Data Grid
Đã khởi tạo dự án React và TypeScript.
console.log(filterData(sample, "Viet", columns));
// [] – vẫn phân biệt dấu
console.log(filterData(sample, "3.5", columns).map(s => s.id));
// ["SV01", "SV03"]
console.log(filterData(sample, "", columns).map(s => s.id));
// ["SV01", "SV02", "SV03", "SV04"]
console.log(filterData(sample, " ", columns).map(s => s.id));
// ["SV01", "SV02", "SV03", "SV04"]
console.log(filterData(sample, " Minh Anh ", columns).map(s => s.id));
// ["SV02"]
console.log(filterData(sample, "không tồn tại", columns));
// []
console.log(filterData([], "Việt", columns));
// []
console.log(filterData(sample, "Việt", []));
// []
console.log(filterData(sample, "", []).length);
// 4 – query rỗng ưu tiên trả bản sao toàn bộ data
console.log(filterData(sample, "SV01", columns));
// [] – id không nằm trong columns
(2) ['SV01', 'SV03']
VM49:1
Relevant data is sent to Google

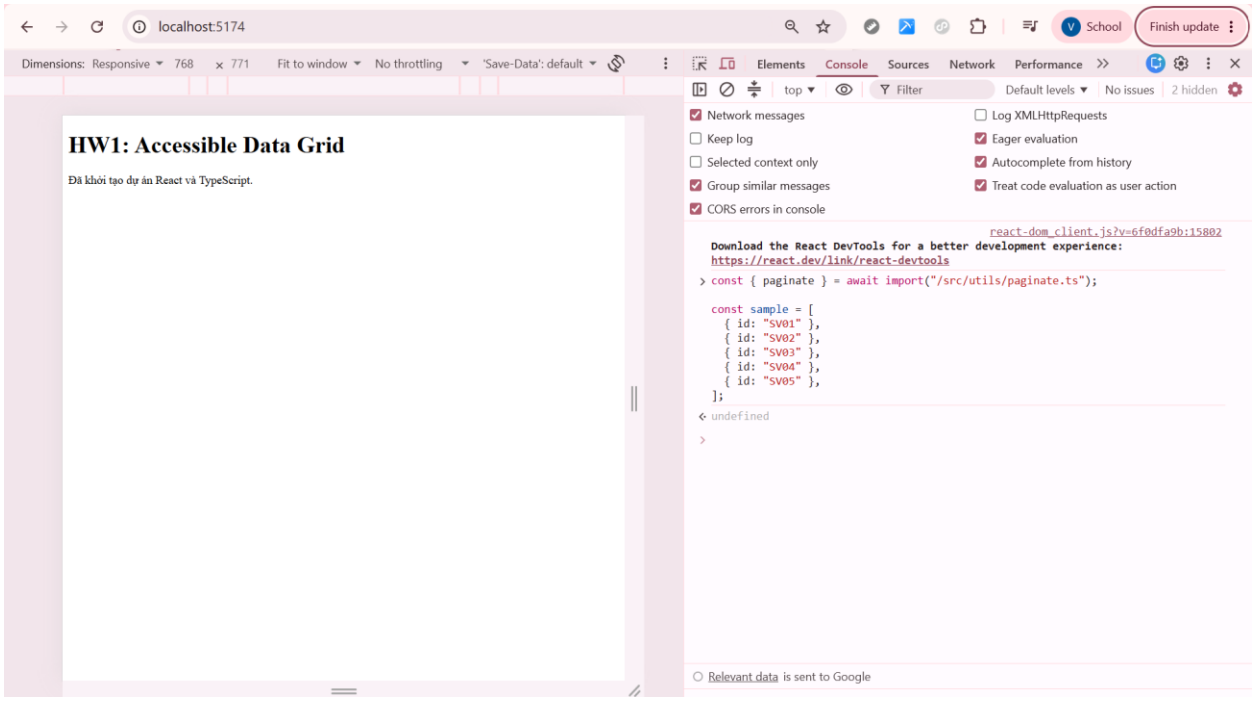

Dimensions: Responsive 768 x 771 Fit to window No throttling 'Save-Data': default
HW1: Accessible Data Grid
Đã khởi tạo dự án React và TypeScript.
Elements Console Sources Network Performance
top Filter Default levels No issues 2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
// [] – id không nằm trong columns
(2) ['SV01', 'SV03'] VM49:1
(2) ['SV01', 'SV03'] VM49:4
[] VM49:7
(2) ['SV01', 'SV03'] VM49:10
(4) ['SV01', 'SV02', 'SV03', 'SV04'] VM49:13
(4) ['SV01', 'SV02', 'SV03', 'SV04'] VM49:16
['SV02'] VM49:19
[] VM49:22
[] VM49:25
[] VM49:28
4 VM49:31
[] VM49:34
undefined
const mixed = [
{ value: null },
{ value: undefined },
{ value: true },
{ value: { name: "Việt" }
{ value: "Việt" },
{ value: 3.5 },
];
Công cụ cắt
Đã sao chép ảnh chụp màn hình vào bảng tạm
Tự động lưu vào thư mục ảnh chụp màn hình.
Đánh dấu và Chia sẻ
Relevant data is sent to Google

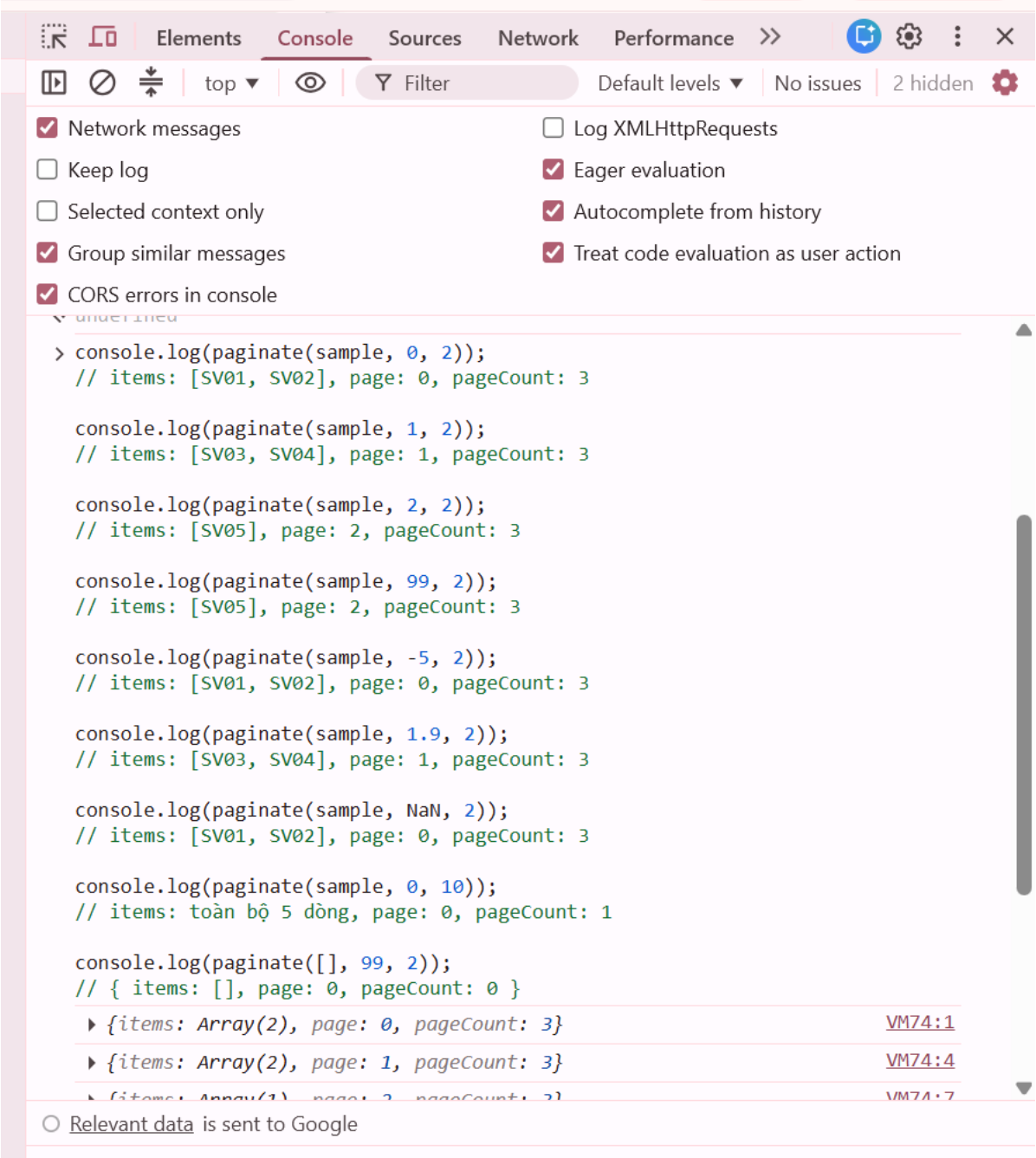

Dimensions: Responsive 768 x 771 Fit to window No throttling 'Save-Data': default
HW1: Accessible Data Grid
Đã khởi tạo dự án React và TypeScript.
Elements Console Sources Network Performance
top Filter Default levels No issues 2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
const mixed = [
{ value: null },
{ value: undefined },
{ value: true },
{ value: { name: "Việt" } },
{ value: "Việt" },
{ value: 3.5 },
];
const valueColumn = [
{ header: "Giá trị", accessorKey: "value" },
];
console.log(filterData(mixed, "Việt", valueColumn));
// [{ value: "Việt" }]
console.log(filterData(mixed, "true", valueColumn));
// []
[{…}] VM53:14
[] VM53:17
undefined
Object.freeze(input);
Công cụ cắt
Đã sao chép ảnh chụp màn hình vào bảng tạm
Tự động lưu vào thư mục ảnh chụp màn hình.
Đánh dấu và Chia sẻ
Relevant data is sent to Google





2.3.paginate(data, page, size)


localhost:5174
School
Finish update
Dimensions: Responsive 768 x 771 Fit to window No throttling 'Save-Data': default
Elements Console Sources Network Performance
top Filter Default levels No issues 2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
HW1: Accessible Data Grid
Đã khởi tạo dự án React và TypeScript.
react-dom_client.js?v=6f0dfa9b:15802
Download the React DevTools for a better development experience:
https://react.dev/link/react-devtools
const { paginate } = await import("/src/utils/paginate.ts");
const sample = [
{ id: "SV01" },
{ id: "SV02" },
{ id: "SV03" },
{ id: "SV04" },
{ id: "SV05" },
];
undefined
Relevant data is sent to Google


Elements
Console
Sources
Network
Performance
top
Filter
Default levels
No issues
2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
undefined
> console.log(paginate(sample, 0, 2));
// items: [SV01, SV02], page: 0, pageCount: 3
console.log(paginate(sample, 1, 2));
// items: [SV03, SV04], page: 1, pageCount: 3
console.log(paginate(sample, 2, 2));
// items: [SV05], page: 2, pageCount: 3
console.log(paginate(sample, 99, 2));
// items: [SV05], page: 2, pageCount: 3
console.log(paginate(sample, -5, 2));
// items: [SV01, SV02], page: 0, pageCount: 3
console.log(paginate(sample, 1.9, 2));
// items: [SV03, SV04], page: 1, pageCount: 3
console.log(paginate(sample, NaN, 2));
// items: [SV01, SV02], page: 0, pageCount: 3
console.log(paginate(sample, 0, 10));
// items: toàn bộ 5 dòng, page: 0, pageCount: 1
console.log(paginate([], 99, 2));
// { items: [], page: 0, pageCount: 0 }
{items: Array(2), page: 0, pageCount: 3}
VM74:1
{items: Array(2), page: 1, pageCount: 3}
VM74:4
Relevant data is sent to Google

```
// items: [SV01, SV02], page: 0, pageCount: 3

console.log(paginate(sample, 1.9, 2));
// items: [SV03, SV04], page: 1, pageCount: 3

console.log(paginate(sample, NaN, 2));
// items: [SV01, SV02], page: 0, pageCount: 3

console.log(paginate(sample, 0, 10));
// items: toàn bộ 5 dòng, page: 0, pageCount: 1

console.log(paginate([], 99, 2));
// { items: [], page: 0, pageCount: 0 }
```

```
▸ {items: Array(2), page: 0, pageCount: 3}                    VM74:1
▸ {items: Array(2), page: 1, pageCount: 3}                    VM74:4
▸ {items: Array(1), page: 2, pageCount: 3}                    VM74:7
▸ {items: Array(1), page: 2, pageCount: 3}                    VM74:10
▸ {items: Array(2), page: 0, pageCount: 3}                    VM74:13
▸ {items: Array(2), page: 1, pageCount: 3}                    VM74:16
▸ {items: Array(2), page: 0, pageCount: 3}                    VM74:19
▸ {items: Array(5), page: 0, pageCount: 1}                    VM74:22
▸ {items: Array(0), page: 0, pageCount: 0}                    VM74:25
```

```
undefined
```

Relevant data is sent to Google

```
for (const size of [0, -1, 1.5, NaN, Infinity]) {
  try {
    paginate(sample, 0, size);
    console.log("Không đạt:", size);
  } catch (error) {
    console.log(size, error instanceof RangeError);
  }
}
// Mỗi trường hợp đều in true.
```

```
0 true                                                        VM78:6
-1 true                                                       VM78:6
1.5 true                                                      VM78:6
NaN true                                                      VM78:6
Infinity true                                                 VM78:6
```

```
undefined
```

Relevant data is sent to Google

```
Object.freeze(input);

const before = JSON.stringify(input);
const result = paginate(input, 1, 2);

console.log(result.items !== input);
// true – mảng mới

console.log(JSON.stringify(input) === before);
// true – đầu vào không đổi

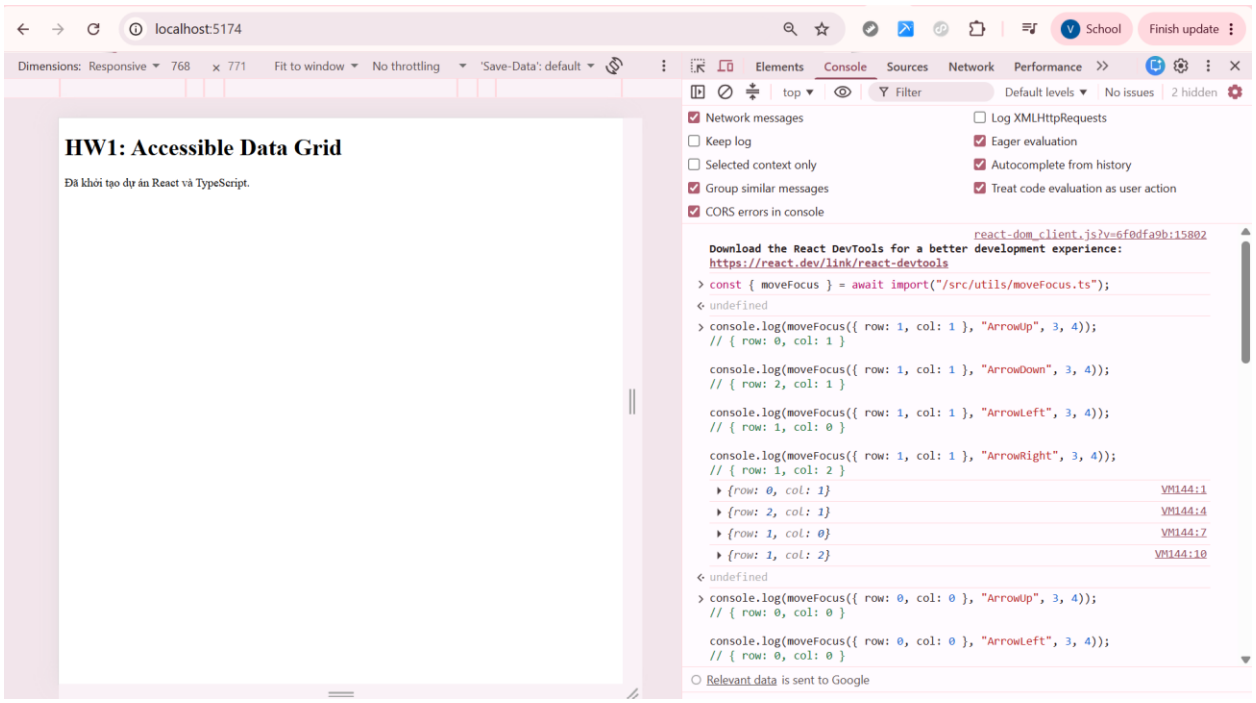
console.log(
  result.items[0] === input[2] &&
  result.items[1] === input[3],
);
// true – giữ tham chiếu object và thứ tự

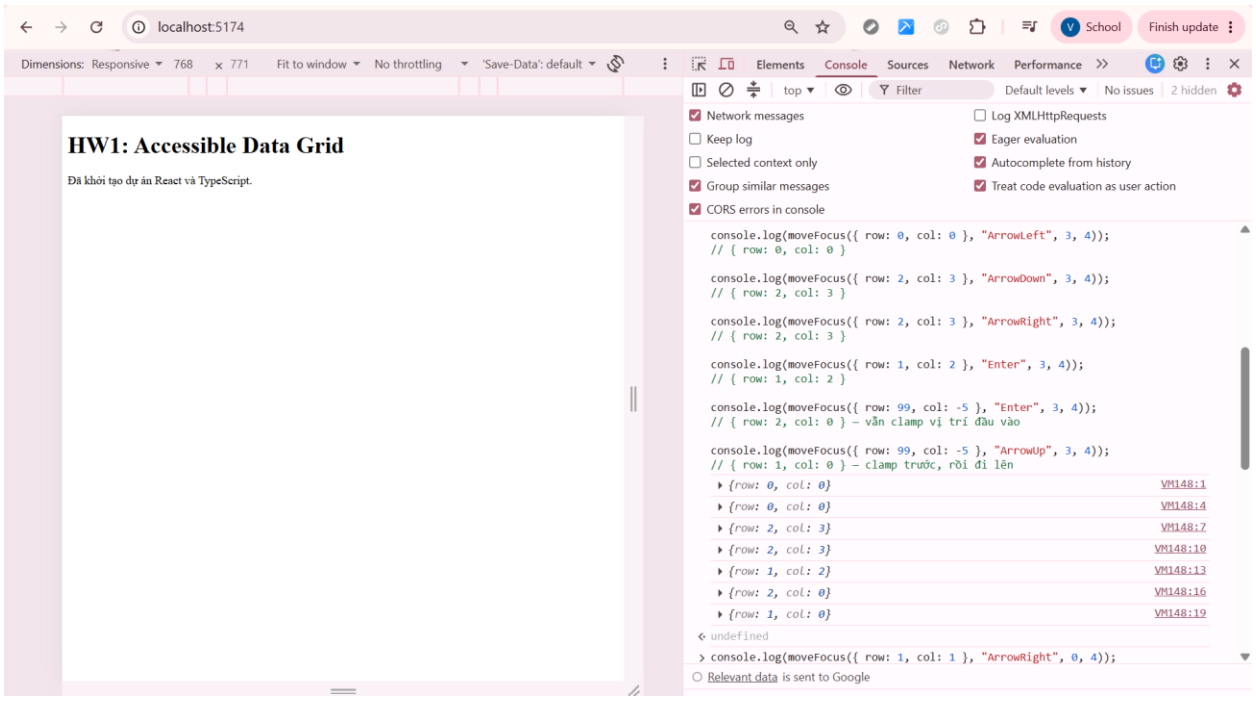
const empty = [];
console.log(paginate(empty, 0, 2).items !== empty);
// true – ngay cả mảng rỗng cũng trả mảng mới
```

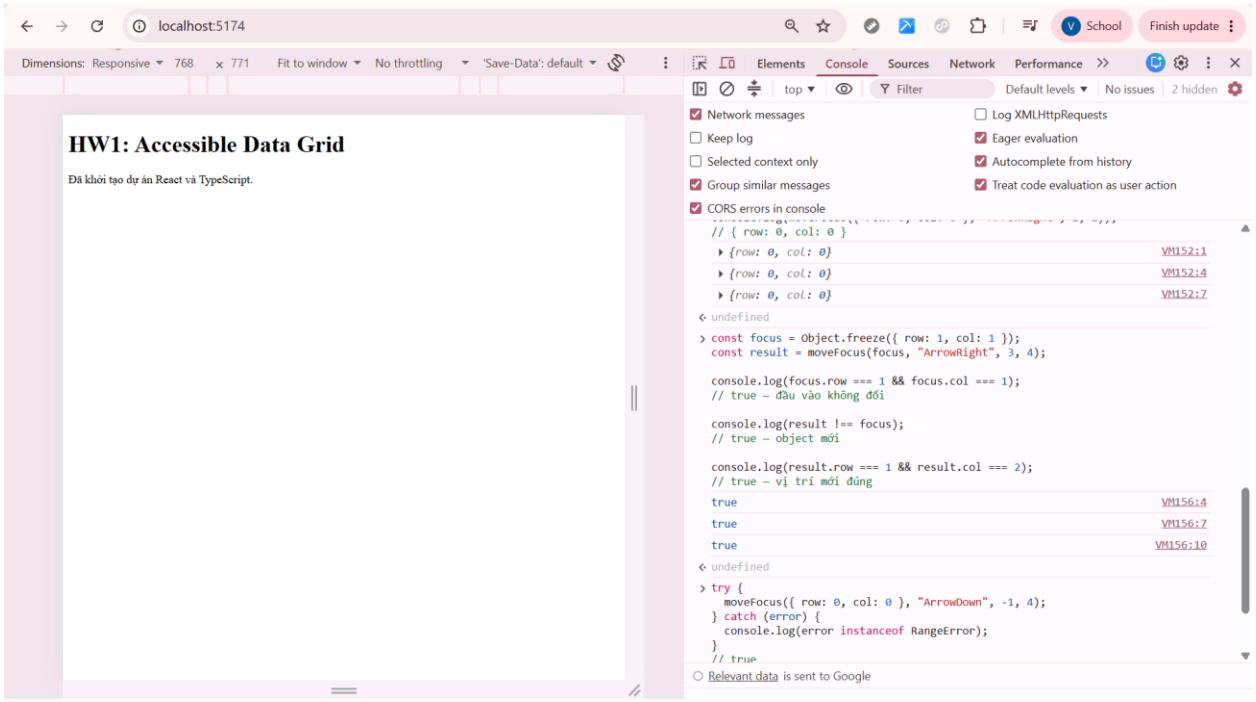
```
true                                                    VM82:10
true                                                    VM82:13
true                                                    VM82:16
true                                                    VM82:23
undefined
>
```

Relevant data is sent to Google

2.4.moveFocus(focus, key, rows, cols)


localhost:5174
School
Finish update
Dimensions: Responsive 768 x 771 Fit to window No throttling 'Save-Data': default
Elements Console Sources Network Performance
top Filter Default levels No issues 2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
HW1: Accessible Data Grid
Đã khởi tạo dự án React và TypeScript.
react-dom_client.js?v=6f0dfa9b:15802
Download the React DevTools for a better development experience: https://react.dev/link/react-devtools
const { moveFocus } = await import("/src/utils/moveFocus.ts");
undefined
console.log(moveFocus({ row: 1, col: 1 }, "ArrowUp", 3, 4));
// { row: 0, col: 1 }
console.log(moveFocus({ row: 1, col: 1 }, "ArrowDown", 3, 4));
// { row: 2, col: 1 }
console.log(moveFocus({ row: 1, col: 1 }, "ArrowLeft", 3, 4));
// { row: 1, col: 0 }
console.log(moveFocus({ row: 1, col: 1 }, "ArrowRight", 3, 4));
// { row: 1, col: 2 }
{row: 0, col: 1} VM144:1
{row: 2, col: 1} VM144:4
{row: 1, col: 0} VM144:7
{row: 1, col: 2} VM144:10
undefined
console.log(moveFocus({ row: 0, col: 0 }, "ArrowUp", 3, 4));
// { row: 0, col: 0 }
console.log(moveFocus({ row: 0, col: 0 }, "ArrowLeft", 3, 4));
// { row: 0, col: 0 }
Relevant data is sent to Google


localhost:5174
School
Finish update
Dimensions: Responsive 768 x 771 Fit to window No throttling 'Save-Data': default
Elements Console Sources Network Performance
top Filter Default levels No issues 2 hidden
Network messages
Keep log
Selected context only
Group similar messages
CORS errors in console
Log XMLHttpRequests
Eager evaluation
Autocomplete from history
Treat code evaluation as user action
HW1: Accessible Data Grid
Đã khởi tạo dự án React và TypeScript.
console.log(moveFocus({ row: 0, col: 0 }, "ArrowLeft", 3, 4));
// { row: 0, col: 0 }
console.log(moveFocus({ row: 2, col: 3 }, "ArrowDown", 3, 4));
// { row: 2, col: 3 }
console.log(moveFocus({ row: 2, col: 3 }, "ArrowRight", 3, 4));
// { row: 2, col: 3 }
console.log(moveFocus({ row: 1, col: 2 }, "Enter", 3, 4));
// { row: 1, col: 2 }
console.log(moveFocus({ row: 99, col: -5 }, "Enter", 3, 4));
// { row: 2, col: 0 } – vẫn clamp vị trí đầu vào
console.log(moveFocus({ row: 99, col: -5 }, "ArrowUp", 3, 4));
// { row: 1, col: 0 } – clamp trước, rồi đi lên
{row: 0, col: 0} VM148:1
{row: 0, col: 0} VM148:4
{row: 2, col: 3} VM148:7
{row: 2, col: 3} VM148:10
{row: 1, col: 2} VM148:13
{row: 2, col: 0} VM148:16
{row: 1, col: 0} VM148:19
undefined
console.log(moveFocus({ row: 1, col: 1 }, "ArrowRight", 0, 4));
Relevant data is sent to Google




```
console.log(result.row === 1 && result.col === 2);
// true – vị trí mới đúng
```

true VM156:4

true VM156:7

true VM156:10

undefined

```
try {
  moveFocus({ row: 0, col: 0 }, "ArrowDown", -1, 4);
} catch (error) {
  console.log(error instanceof RangeError);
}
// true
```

true VM160:4

undefined

[vite] server connection lost. Polling for restart... client:1090

Relevant data is sent to Google